



Image: Mikael Häggström, CC0.

A CT scan of a brain.

It only counted X-rays.

Last lesson: climb.

The score is the **likelihood** of your observed data given the parameters.



By the end, you will be able to:

1. Write the likelihood of your data given the parameters

2. Use a computer to get the log likelihood and its derivative

3. Find the maximum likelihood estimate by climbing

4. Climb many parameters at once, as a CT scanner does



Review





$$\mathrm{P}(A \text{ and } B) = \mathrm{P}(A) \cdot \mathrm{P}(B)$$

independent: multiply

Last lesson



$$\text{new } \theta = \theta + \text{step size} \times \frac{\partial \, \text{score}}{\partial \theta}$$



A formula: $\partial \, \text{score} / \partial \, \theta$ in one look



Three ways to write small numbers

Decimal	Scientific	$\log_{10}$
0.1	10^{-1}	-1
0.01	10^{-2}	-2
0.001	10^{-3}	-3
$\vdots$	$\vdots$	$\vdots$
0.000244	2.44×10^{-4}	-3.61



End Review



The sky, cut into patches

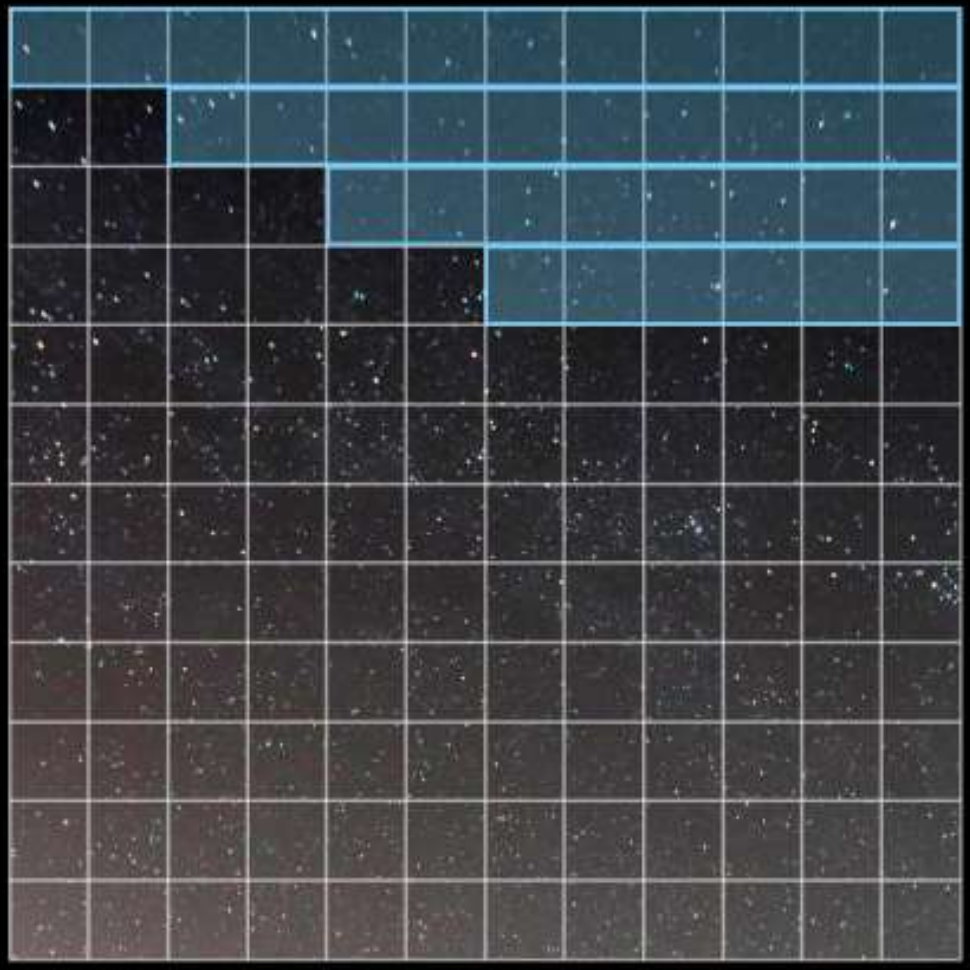
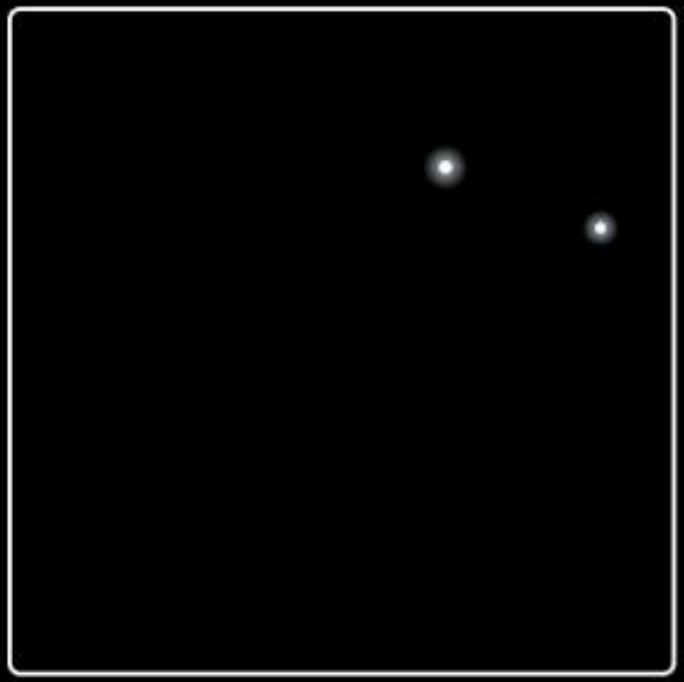


Photo: NASA/KSC, public domain; brightened for display.

today: the dark patches outside the Milky Way

Last week: the Milky Way



2 stars

Stars in a dark patch: a Poisson with average θ

θ 2.0

0 10

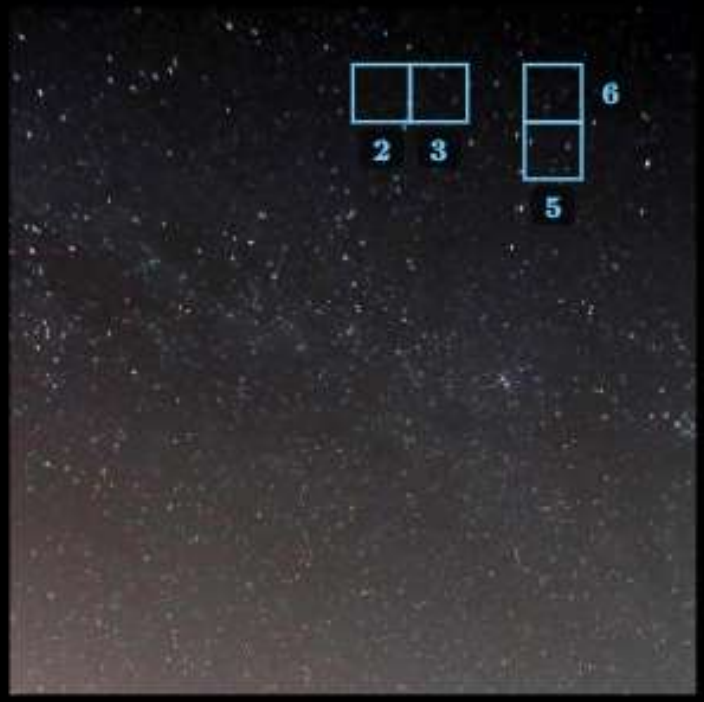
average stars per patch

New sample



Run it backwards

You guess: which θ made these?



the real sky

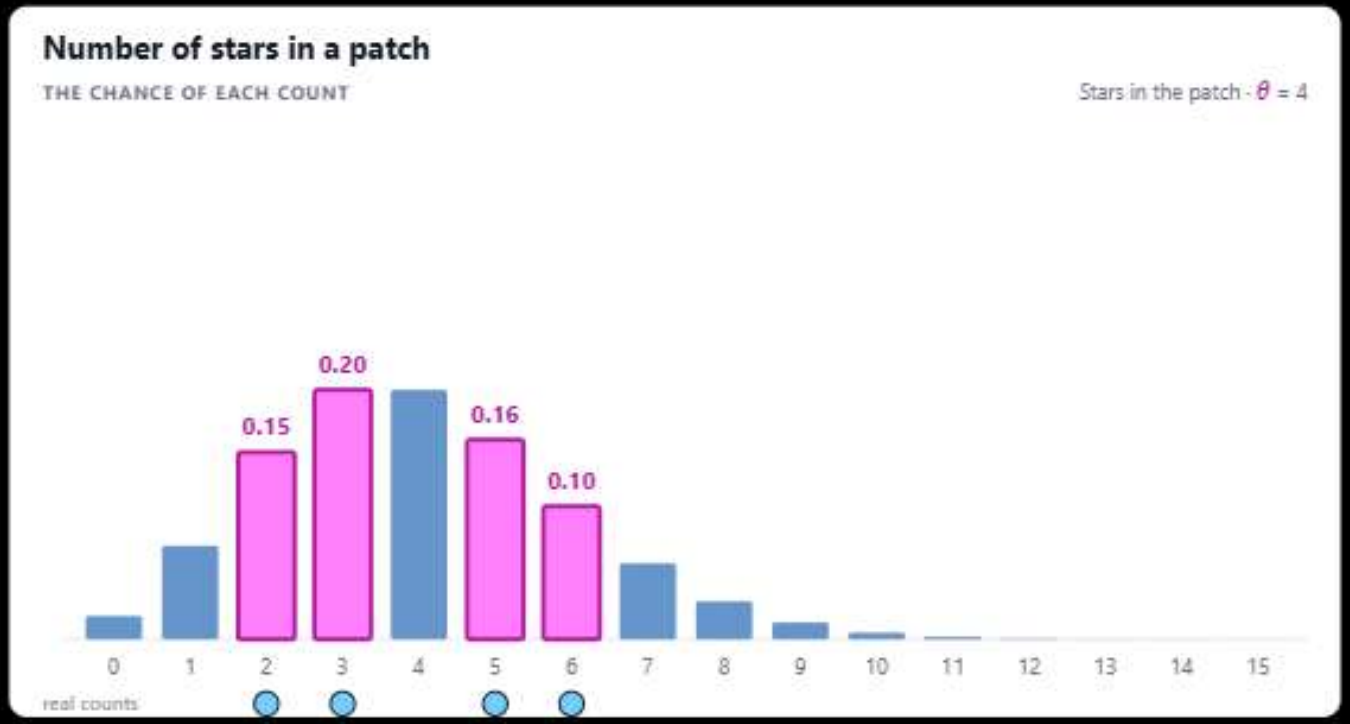
θ: average stars in a dark patch



Stars in the patch

Photo: NASA/KSC, public domain; brightened for display.

The chance of what we saw



θ 4

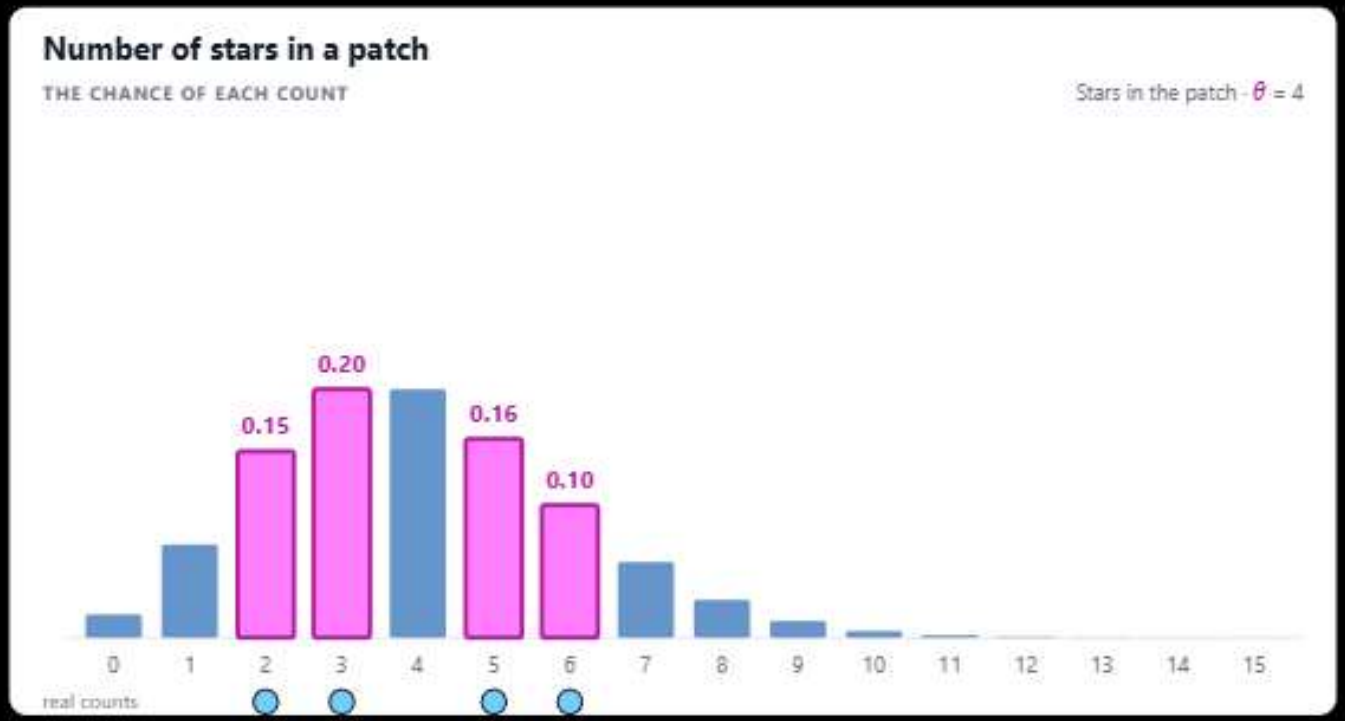
average stars in a dark patch

$\theta - 1$
$\approx$ **0.00000002**

$\theta - 4$
$\approx$ **0.0005**

$0.15 \times 0.20 \times 0.16 \times 0.10 \approx \mathbf{0.0005}$

The chance of what we saw



θ 4

average stars in a dark patch

$\theta = 1$
$\approx$ **0.00000002**

$\theta = 4$
$\approx$ **0.0005**

$\theta = 9$
$\approx$ **0.0000004**

$0.15 \times 0.20 \times 0.16 \times 0.10 \approx \mathbf{0.0005}$



$$\text{likelihood}(\theta) = \mathrm{P}(2 \mid \theta) \cdot \mathrm{P}(3 \mid \theta) \cdot \mathrm{P}(5 \mid \theta) \cdot \mathrm{P}(6 \mid \theta)$$

a good story puts tall bars where the data landed

θ: average stars in a dark patch



How many words does Chris' daughter know?



How many words do you know?

Testing all 61,800 takes days.

Test 20 at random, then estimate.



Source: Brysbaert, Stevens, Mandera & Keuleers (2016), Frontiers in Psychology, CC BY.

Twenty random words

cenotaph ✗	mulishness ✗	tonsured ✗	straightforward ✓
similar ✓	bacchantes ✗	passport ✓	caduceus ✗
keyboardist ✓	shunt ✓	incliner ✓	interdictor ✓
polling ✓	glom ✗	sponginess ✓	cunningness ✓
rainbow ✓	jejune ✗	amortize ✓	unrounded ✓

An example viewer knew **13** of 20



One parameter

You guess: what θ fits these answers best?

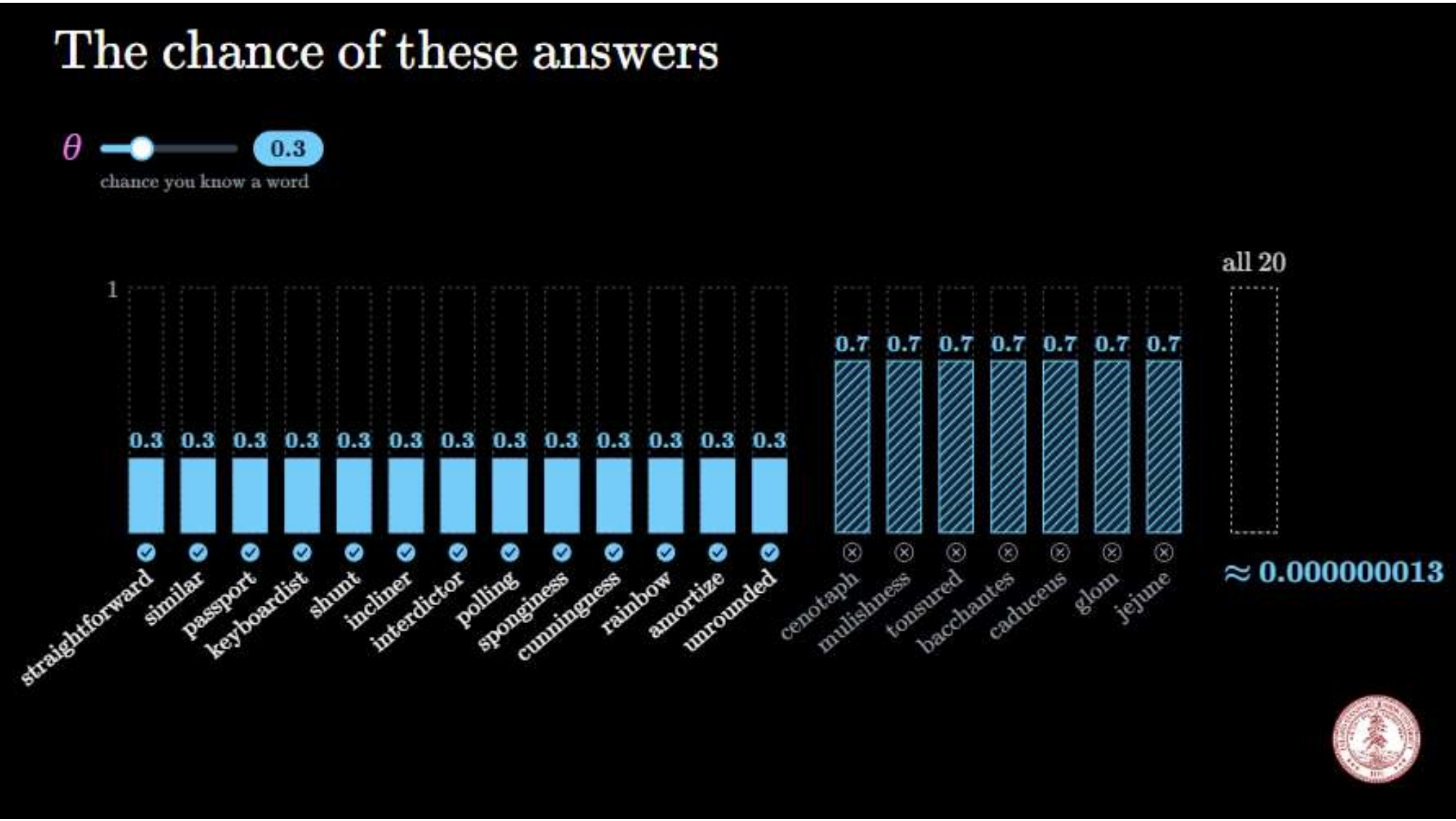
cenotaph ✗	mulishness ✗	tonsured ✗	**straightforward** ✓
similar ✓	bacchantes ✗	**passport** ✓	caduceus ✗
keyboardist ✓	**shunt** ✓	**incliner** ✓	**interdictor** ✓
polling ✓	glom ✗	**sponginess** ✓	**cunningness** ✓
rainbow ✓	jejune ✗	**amortize** ✓	**unrounded** ✓

θ 0 1 ?

chance you know a random word



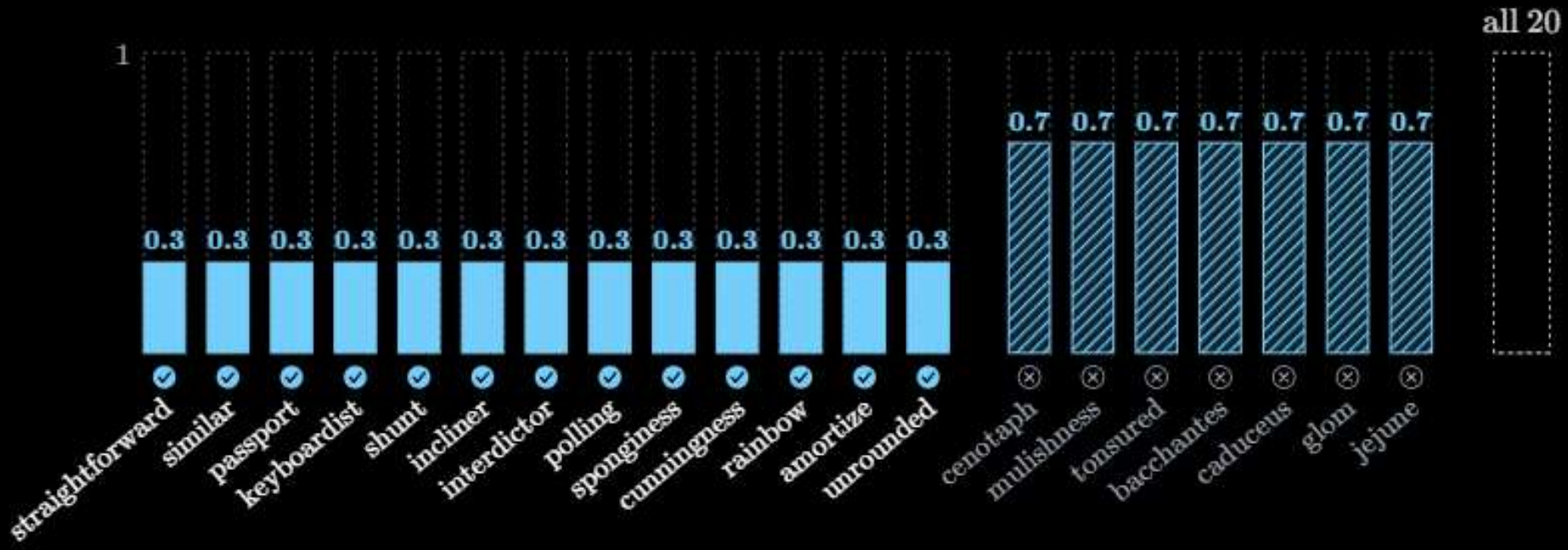
The chance of these answers



The chance of these answers

θ 0.3

chance you know a word



$$0.3^{13} \times 0.7^{7} \approx \mathbf{0.000000013}$$

Three settings of the parameter

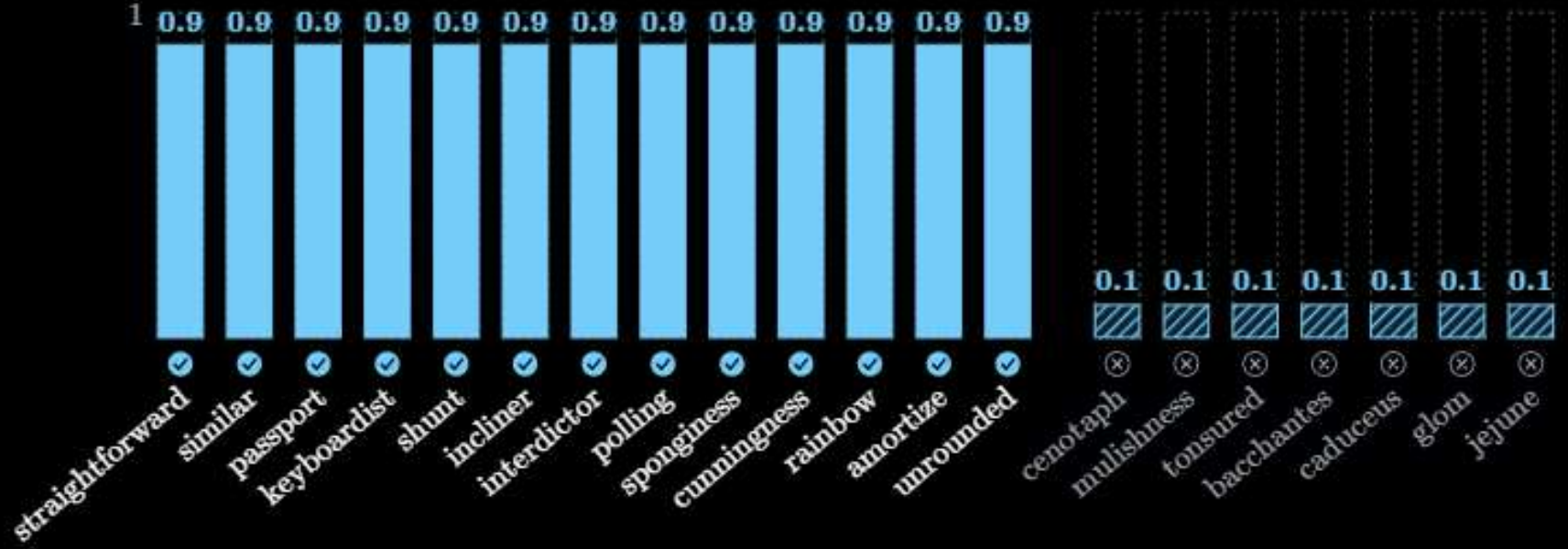
θ — 0.9

chance you know a word

$\theta = 0.3 \quad 0.3^{13} \times 0.7^{7} \approx \mathbf{0.000000013}$

$\theta = 0.5 \quad 0.5^{13} \times 0.5^{7} \approx \mathbf{0.00000095}$

$\theta = 0.9 \quad 0.9^{13} \times 0.1^{7} \approx \mathbf{0.000000025}$





Three settings of the parameter

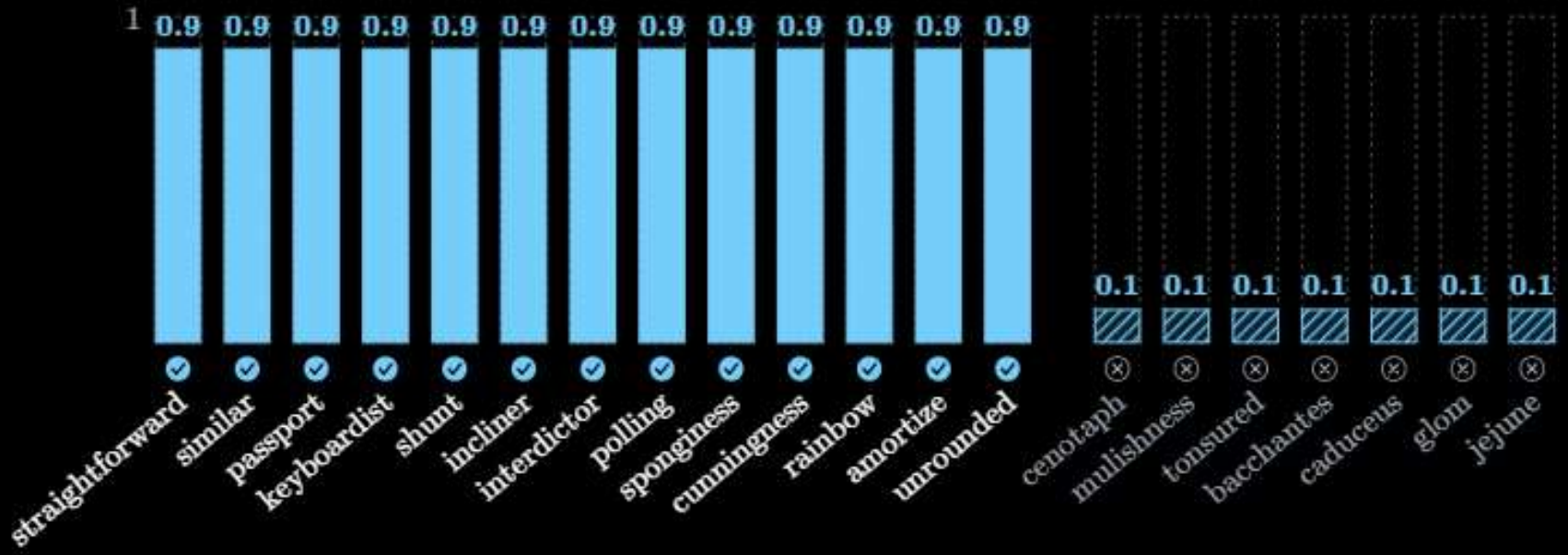
θ — 0.9

chance you know a word

$\theta = 0.3 \quad 0.3^{13} \times 0.7^{7} \approx \mathbf{0.000000013}$

$\theta = 0.5 \quad 0.5^{13} \times 0.5^{7} \approx \mathbf{0.00000095}$

$\theta = 0.9 \quad 0.9^{13} \times 0.1^{7} \approx \mathbf{0.000000025}$



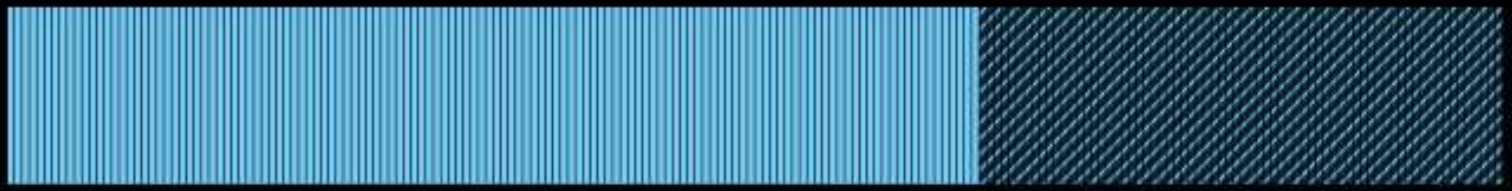


More words, more zeros

θ 0.5

chance you know a word

200 words



$$0.5^{200} = 0.000000000000000000000000000000000000000000000000000000000000$$



Three ways to write small numbers

Decimal	Scientific	$\log_{10}$
0.1	10^{-1}	-1
0.01	10^{-2}	-2
0.001	10^{-3}	-3
$\vdots$	$\vdots$	$\vdots$
0.000244	2.44×10^{-4}	-3.61

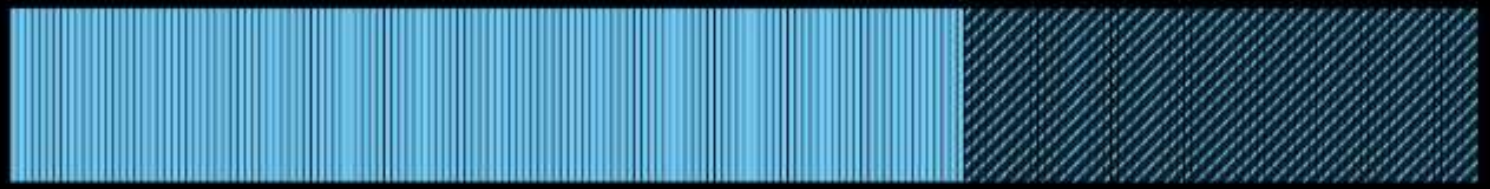


More words, more zeros

θ 0.5

chance you know a word

200 words



$$0.5^{200} = 0.0000000000000000000000000000000000000000000000000000$$

$$\log_{10}(0.5^{200}) \approx -60$$

Let's get formal!





$$\text{likelihood}(\theta) = \text{P}(\text{all data} \mid \theta)$$

$$= \prod_i \text{P}(\text{data}_i \mid \theta)$$



the chance the story gives this very data, read as a score for θ

As a formula

$$\text{Likelihood}(\theta) = \theta^{13}\,(1-\theta)^{7}$$

$$\text{LogLikelihood}(\theta) = 13\log_{10}\theta + 7\log_{10}(1-\theta)$$

$$\text{Likelihood}(0.3) = 0.3^{13} \times 0.7^{7} \approx 0.000000013$$

$$\text{LogLikelihood}(0.3) = 13\log_{10}0.3 + 7\log_{10}0.7 \approx -7.88$$



As a formula

$$\mathrm{P}(\text{answers}) = (\text{chance known})^{13} \times (\text{chance unknown})^{7}$$

$$\mathrm{L}(\theta) = \theta^{13}\,(1-\theta)^{7}$$

$$\mathrm{LL}(\theta) = 13\log_{10}\theta + 7\log_{10}(1-\theta)$$

$$\text{At } \theta = 0.3\text{:} \quad 0.3^{13} \times 0.7^{7} \approx 0.000000013$$



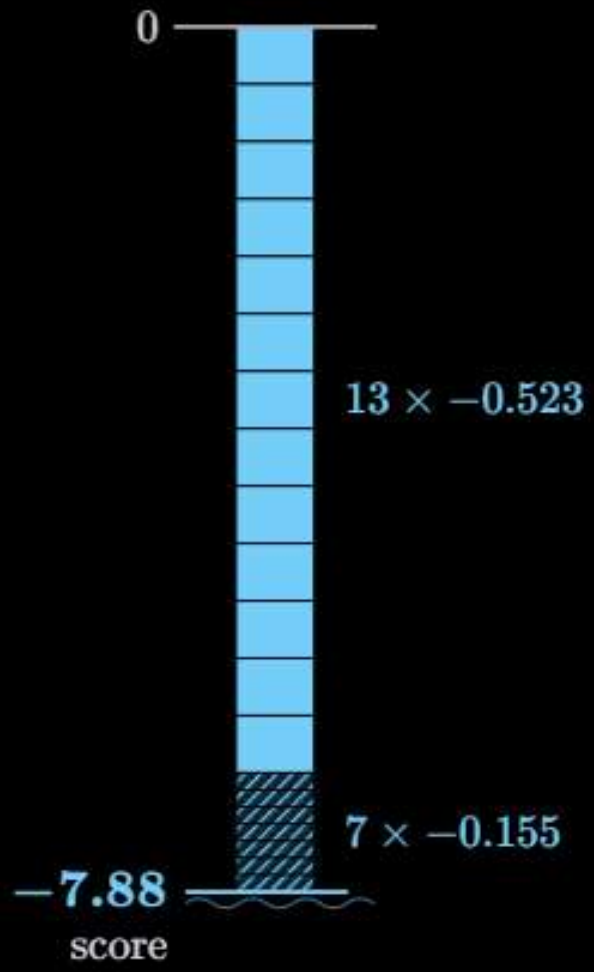


$$\log_{10}(a \cdot b) = \log_{10} a + \log_{10} b$$

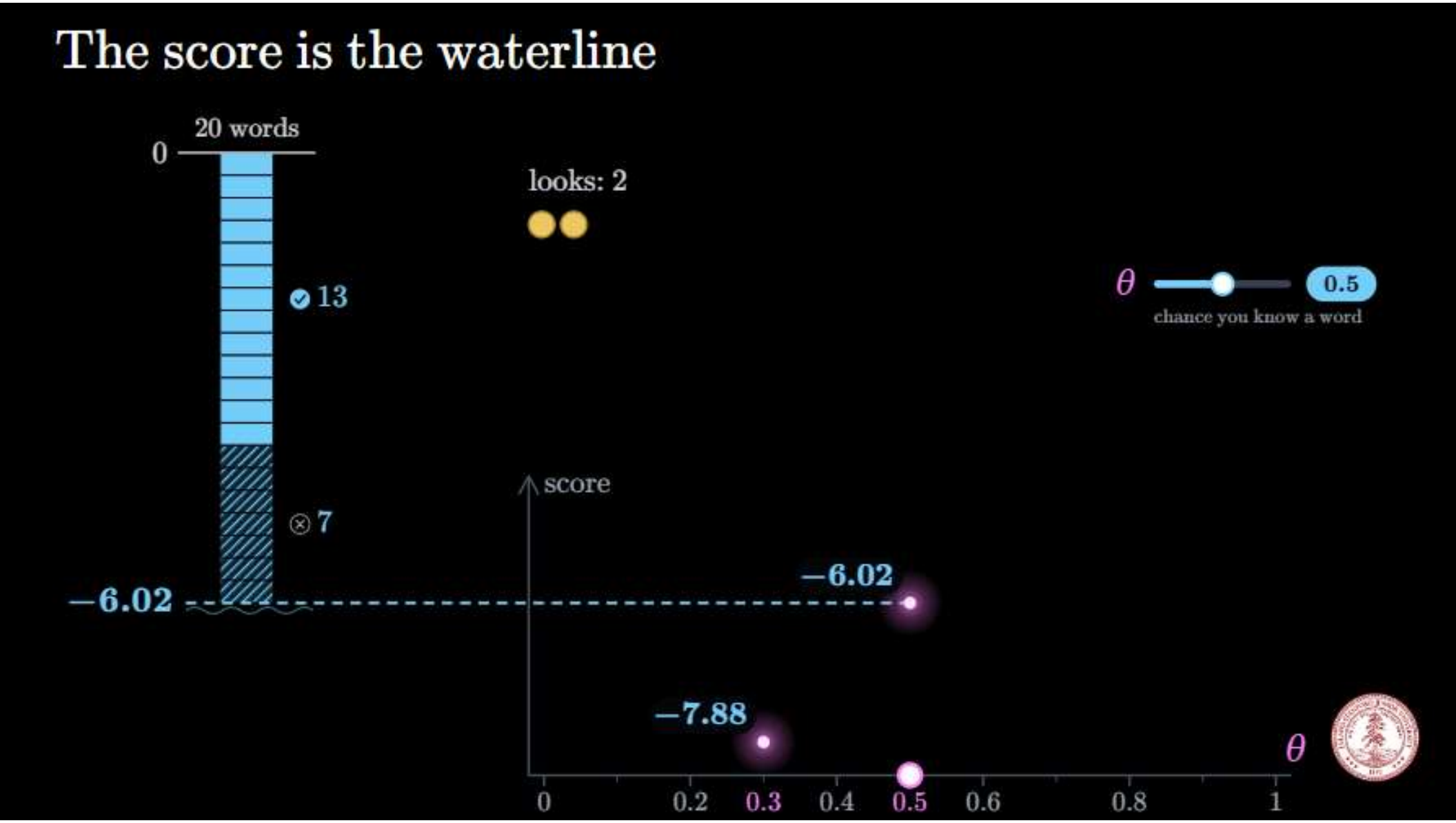
multiply becomes add

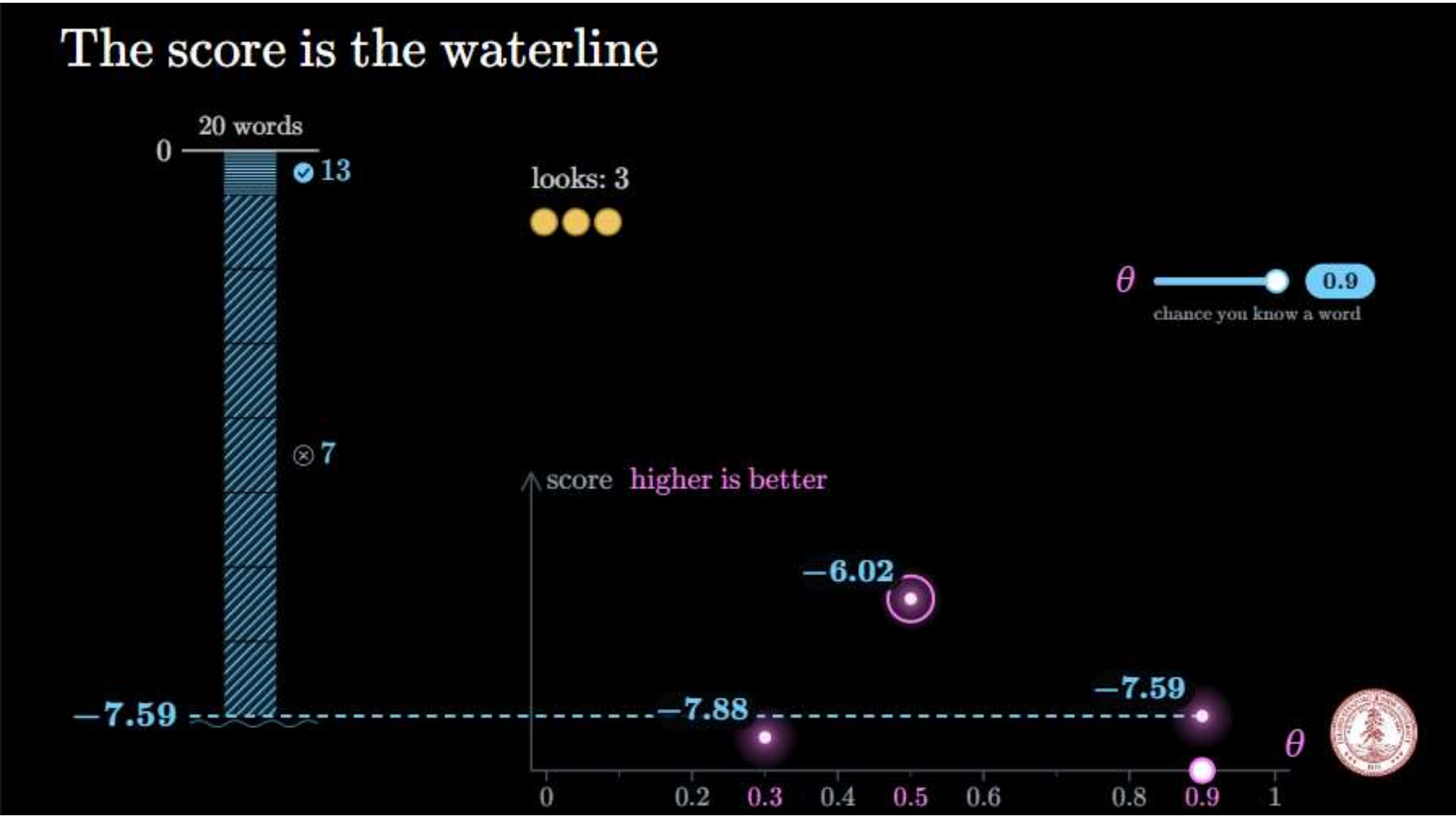


Log likelihood:






The score is the waterline
20 words
0
looks: 2
θ
0.5
chance you know a word
13
7
score
−6.02
−6.02
−7.88
θ
0
0.2
0.3
0.4
0.5
0.6
0.8
1


The score is the waterline
20 words
0
13
7
looks: 3
θ
0.9
chance you know a word
score higher is better
−6.02
−7.59
−7.88
−7.59
θ
0
0.2
0.3
0.4
0.5
0.6
0.8
0.9
1



$$\text{gradient} = \frac{\partial\, \text{score}}{\partial \theta}$$

which way the score rises, and
how fast, per unit of θ

θ: chance you know a word



Code Agent Thinking

Find the slope with respect to θ.



$$\frac{\partial \text{score}}{\partial \theta} = \frac{13}{\theta \ln 10} \cdot 1 + \frac{7}{(1-\theta)\ln 10} \cdot (-1)$$

$$\frac{\partial \text{known}}{\partial \theta} = 1 \qquad \frac{\partial \text{unknown}}{\partial \theta} = -1$$

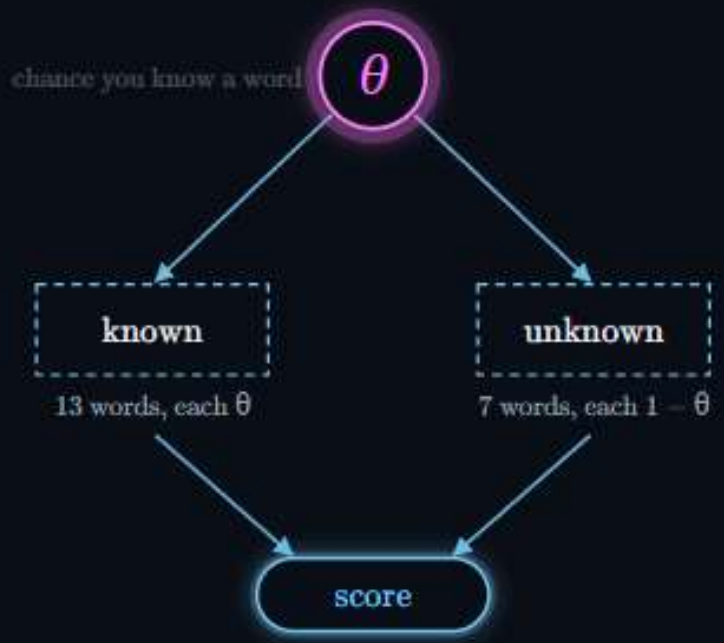
$$\frac{\partial \text{score}}{\partial \text{known}} = \frac{13}{\text{known} \cdot \ln 10}$$

$$\frac{\partial \text{score}}{\partial \text{unknown}} = \frac{7}{\text{unknown} \cdot \ln 10}$$

$$\text{score} = 13 \log_{10} \text{known} + 7 \log_{10} \text{unknown}$$

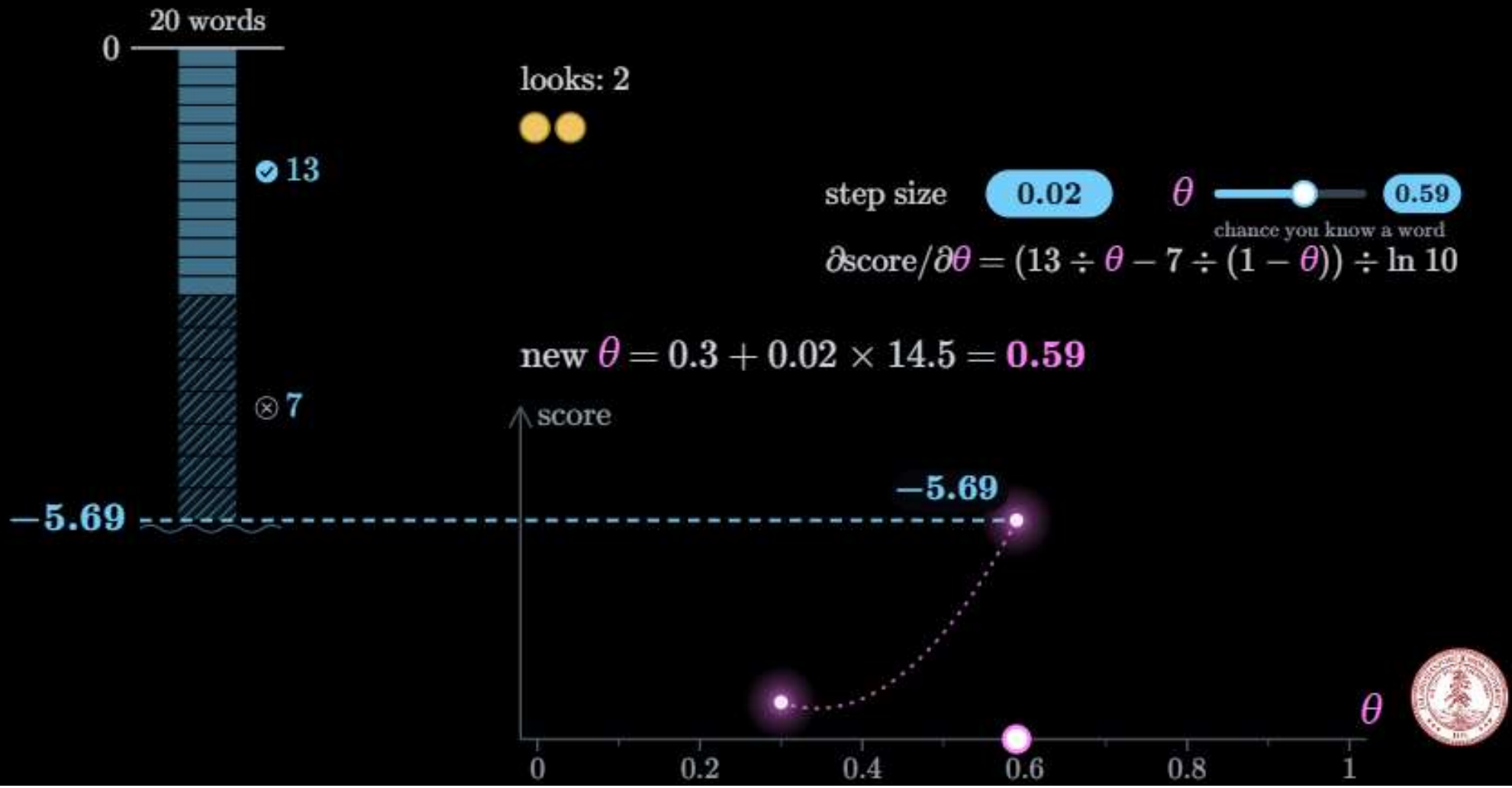
Code Agent ✓ Done

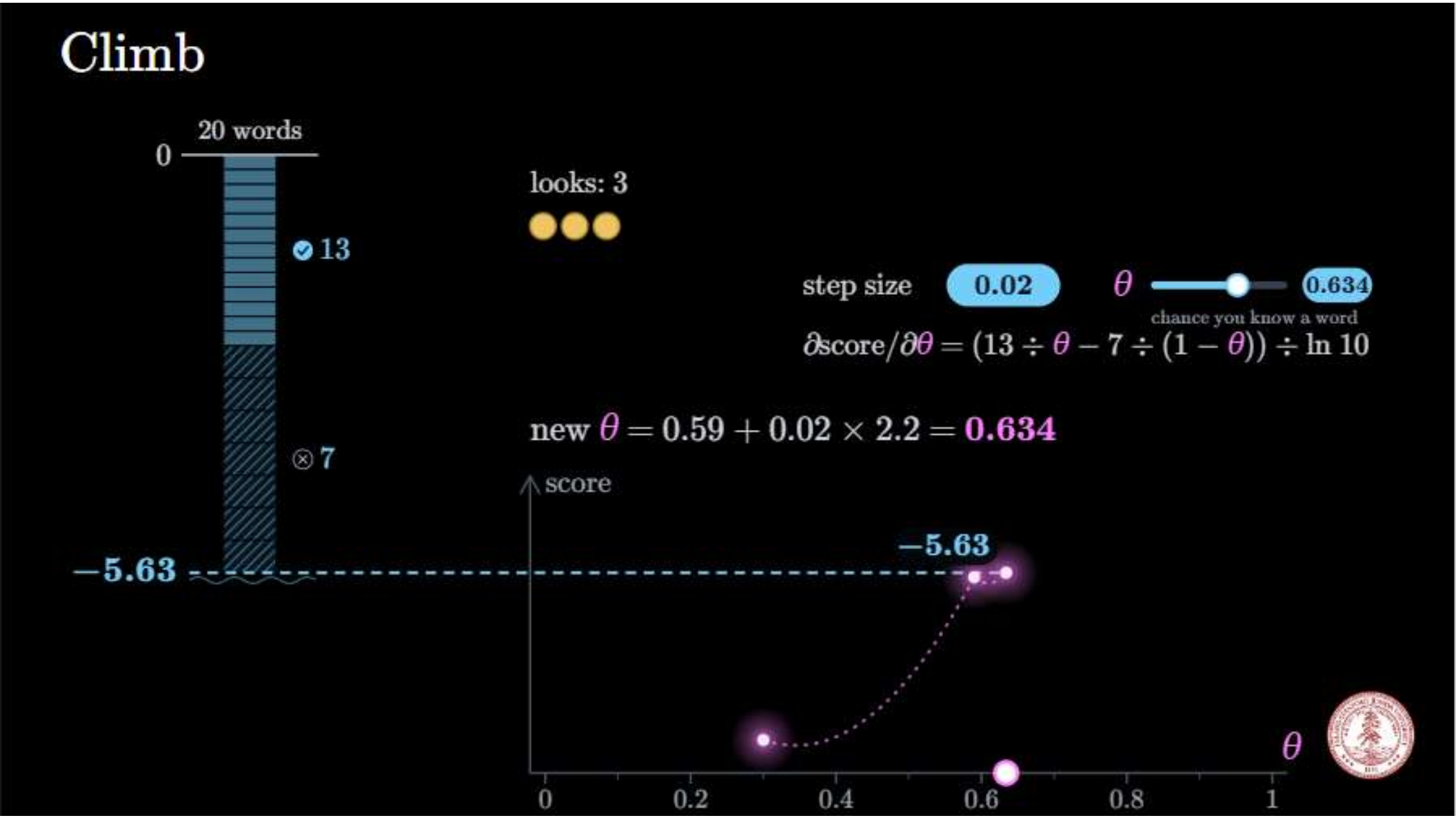
Find the slope with respect to θ.

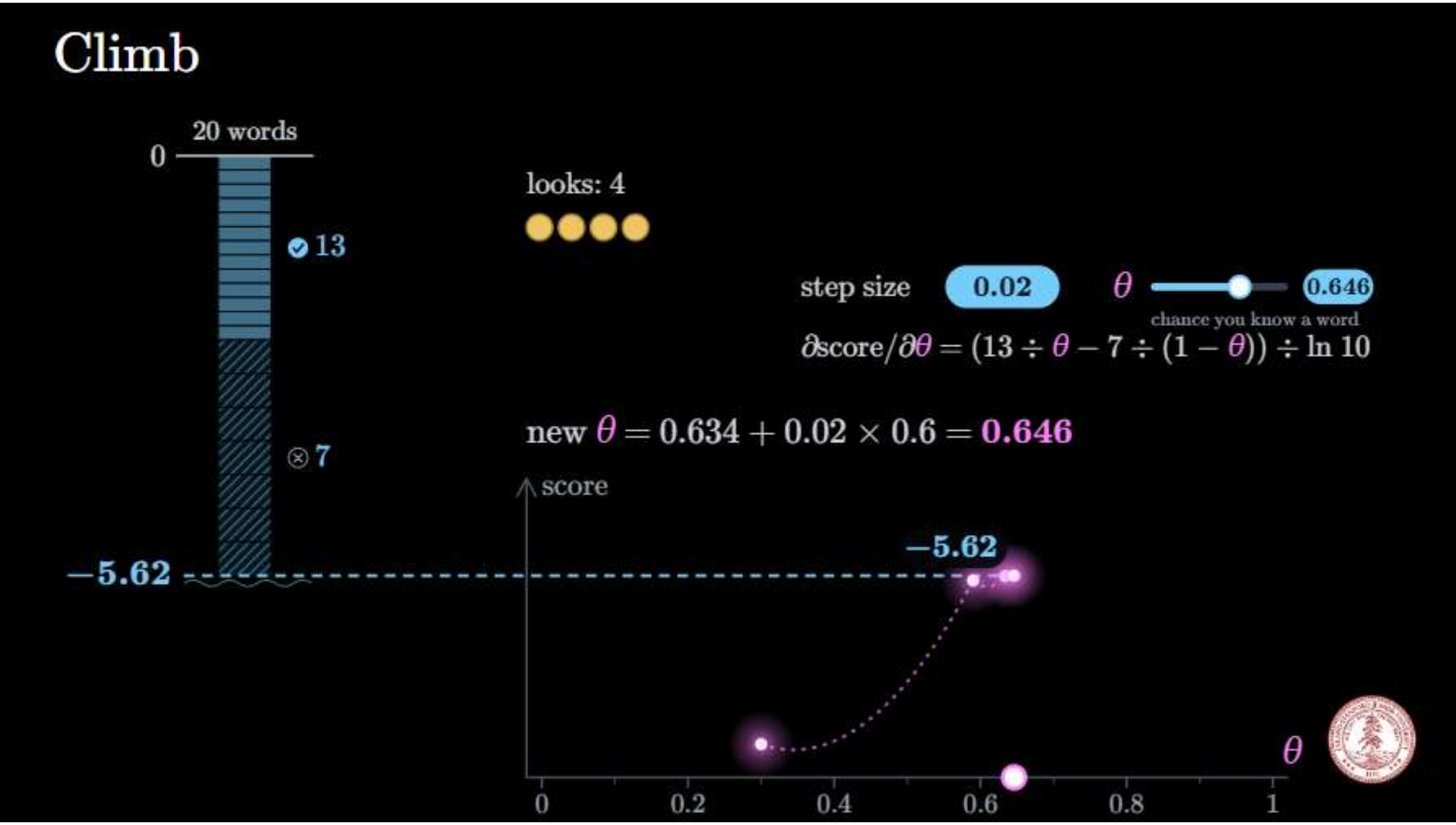


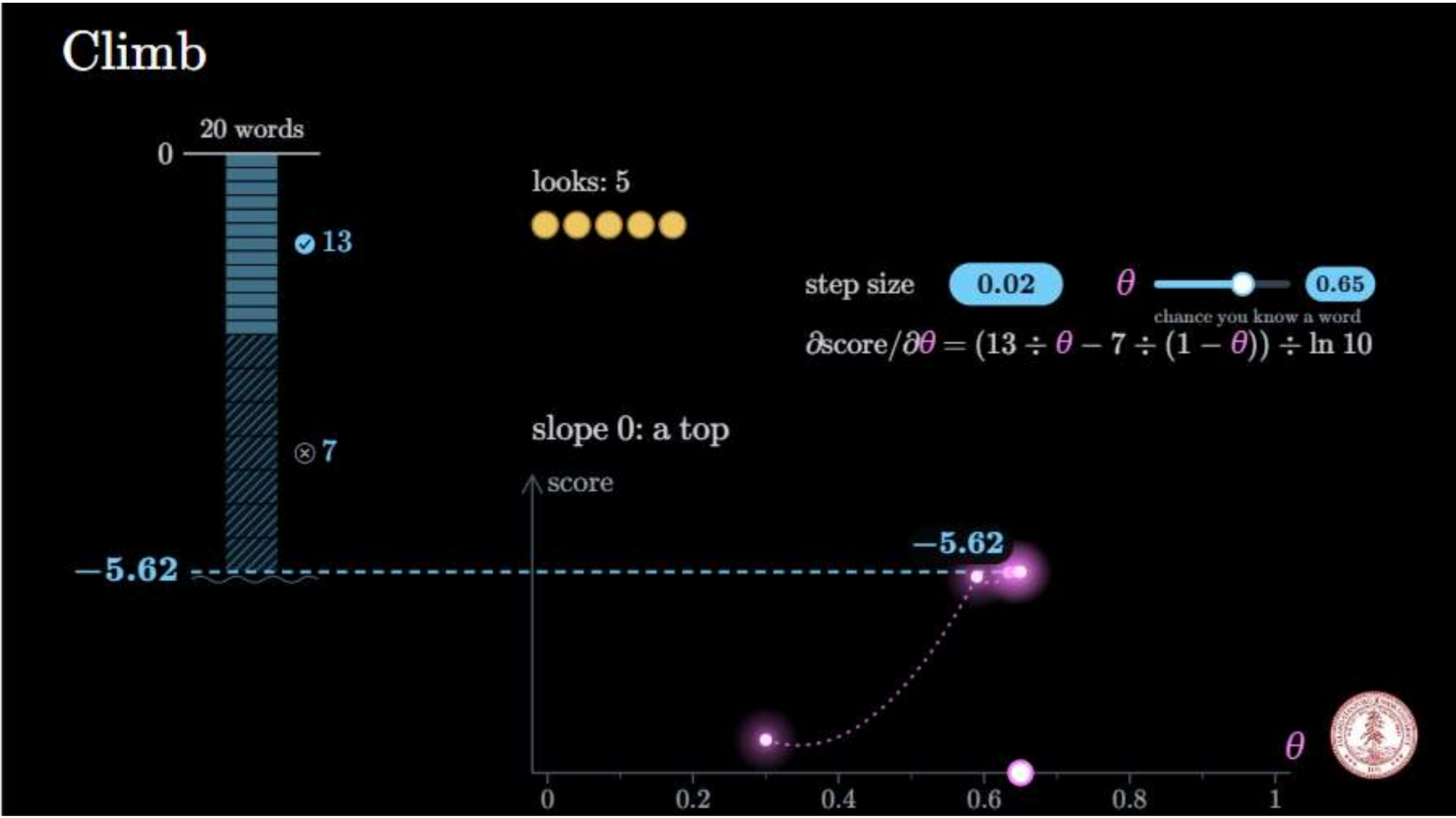
$$\frac{\partial \,\text{score}}{\partial\, \theta} = \frac{1}{\ln 10}\left(\frac{13}{\theta} - \frac{7}{1-\theta}\right)$$

Climb

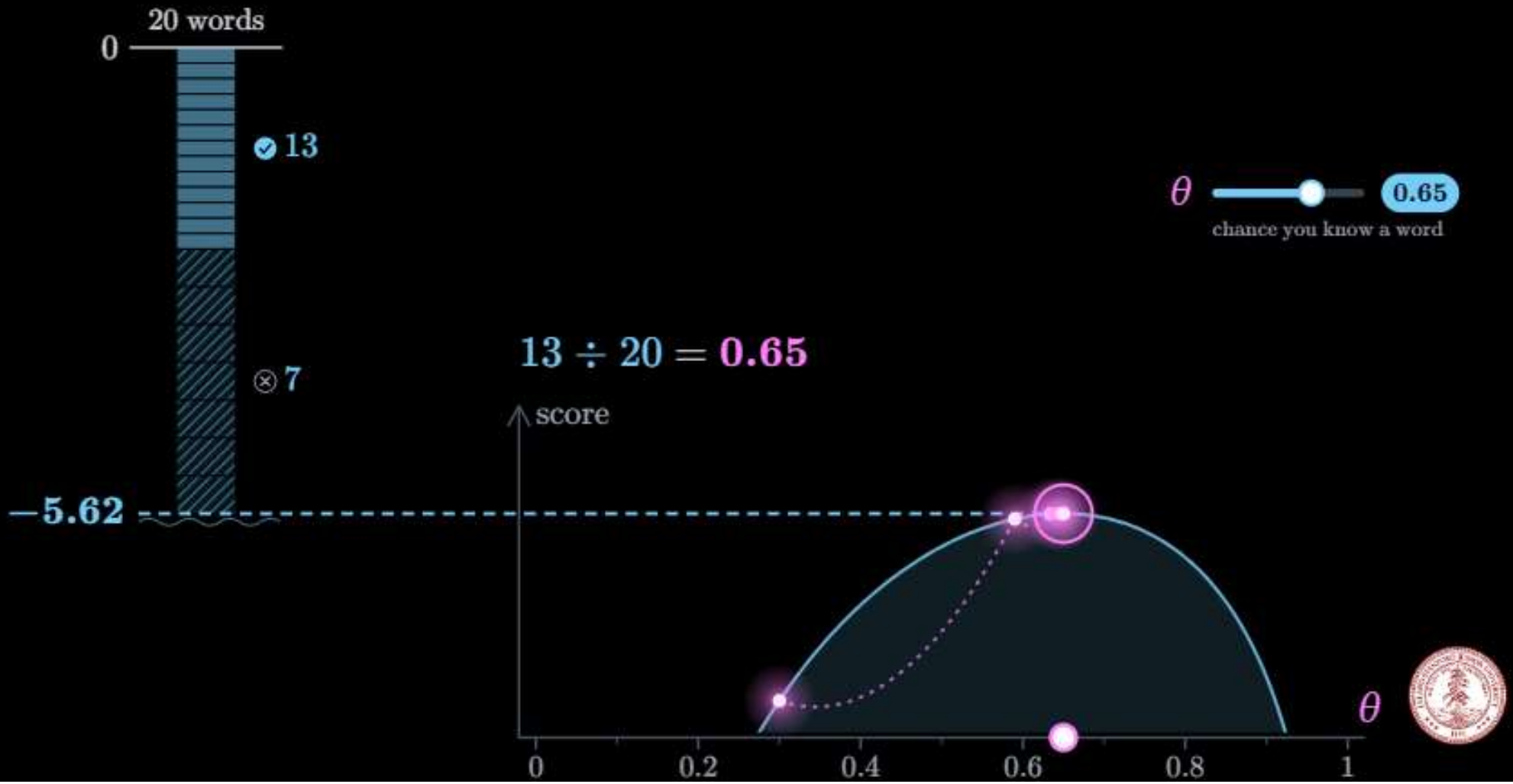

20 words
0
looks: 2
13
7
step size
0.02
θ
0.59
chance you know a word
∂score/∂θ = (13 ÷ θ − 7 ÷ (1 − θ)) ÷ ln 10
new θ = 0.3 + 0.02 × 14.5 = 0.59
score
−5.69
−5.69
θ
0
0.2
0.4
0.6
0.8
1


Climb
20 words
0
13
7
looks: 3
step size
0.02
θ
0.634
chance you know a word
∂score/∂θ = (13 ÷ θ − 7 ÷ (1 − θ)) ÷ ln 10
new θ = 0.59 + 0.02 × 2.2 = 0.634
score
−5.63
−5.63
θ
0
0.2
0.4
0.6
0.8
1

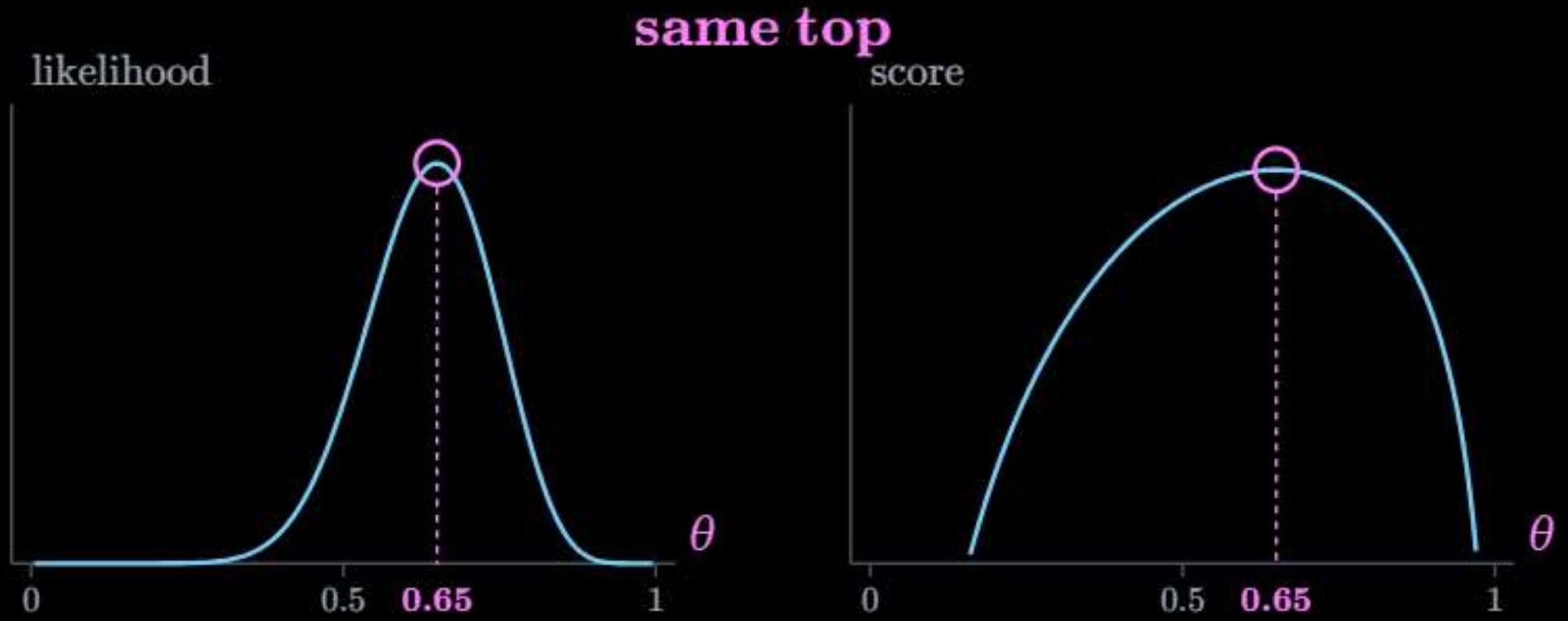

Climb
20 words
0
13
7
looks: 4
step size
0.02
θ
0.646
chance you know a word
∂score/∂θ = (13 ÷ θ − 7 ÷ (1 − θ)) ÷ ln 10
new θ = 0.634 + 0.02 × 0.6 = 0.646
score
−5.62
−5.62
θ
0
0.2
0.4
0.6
0.8
1


Climb
20 words
0
13
7
looks: 5
step size
0.02
θ
0.65
chance you know a word
∂score/∂θ = (13 ÷ θ − 7 ÷ (1 − θ)) ÷ ln 10
slope 0: a top
score
−5.62
−5.62
θ
0
0.2
0.4
0.6
0.8
1

What the climber never saw



Log doesn't change which params win.



θ: chance you know a word

$$\theta_{\text{MLE}} = 13 \div 20 = 0.65$$

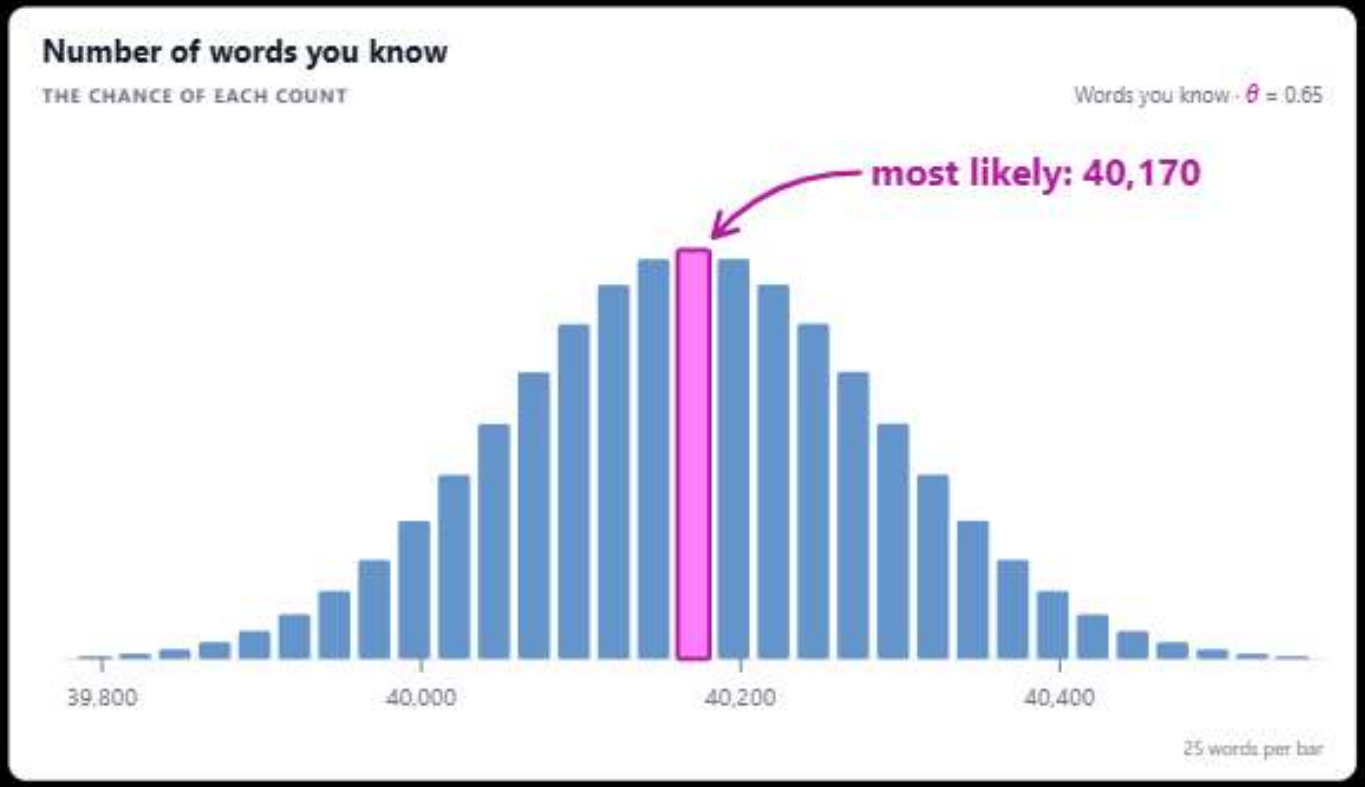
maximum likelihood estimate: the θ that makes the data most likely

θ: chance you know a word



Words you know ~ Binomial(61,800, 0.65)

if θ, your rate, really is 0.65

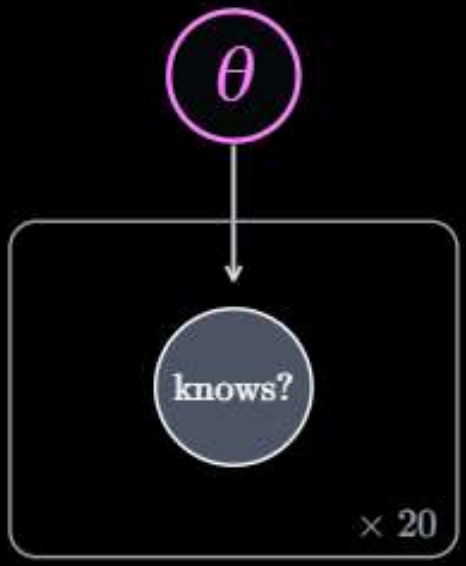


Median 20-year-old native speaker:

42,000

Source: Brysbaert, Stevens, Mandera & Keuleers (2016), Frontiers in Psychology, CC BY (American English).

Step 1: you write the model



For each of 20 words

K **Knows it?**

A Bernoulli, chance θ

θ 0.3

chance you know a random word



Step 2: use a computer for the score

How many words do you know?

Code Agent

What's the score of my answers,
if θ is the chance I know a word?

</> Code updated

Each word adds the log of its chance.
At θ = 0.3: −7.88.

```
def score(θ):
    total = θ
    for knew in answers:
        if knew:
            total += log10(θ)
        else:
            total += log10(1 - θ)
    return total
```

Message Code Agent...





Step 3: use a computer for $\partial \, \mathrm{LL} / \partial \, \theta$

How many words do you know?

Code Agent

```
    return total
```

What's its derivative, ∂score/∂θ?

</> Code updated

Each word you knew adds 1/θ; each you didn't takes away 1/(1 − θ). At θ = 0.3: 14.48.

```
def slope(θ):
    total = 0
    for knew in answers:
        if knew:
            total += 1 / θ
        else:
            total -= 1 / (1 - θ)
    return total / log(10)  # for log10
```

Message Code Agent...





Step 4: use a computer to climb

How many words do you know?

Code Agent

</> Code updated

Done: 5 hops, and it stopped at θ = 0.650.

```
θ = 0.3
step_size = 0.02
tolerance = 0.01
while abs(slope(θ)) > tolerance:
    θ = θ + step_size * slope(θ)
    print(f"θ = {θ:.3f}  score = {score(θ):.2f}")
```

```
θ = 0.590  score = -5.69
θ = 0.633  score = -5.63
θ = 0.646  score = -5.62
θ = 0.649  score = -5.62
θ = 0.650  score = -5.62
```

Message Code Agent…

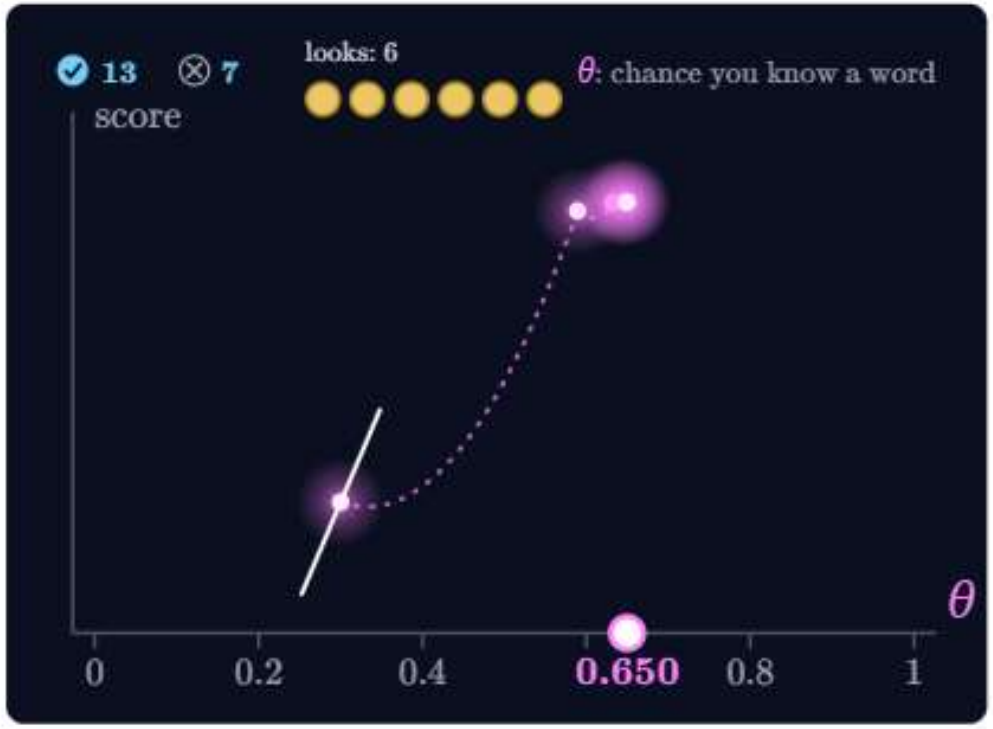


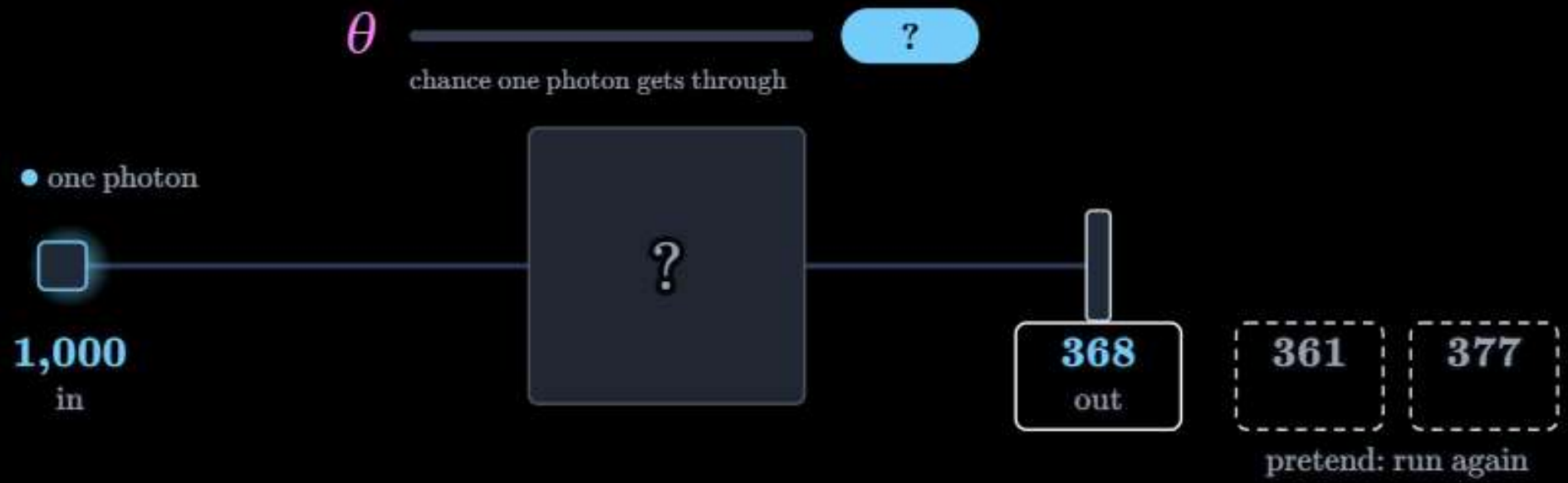


Image: Mikael Häggström, CC0.

A CT scan of a brain.

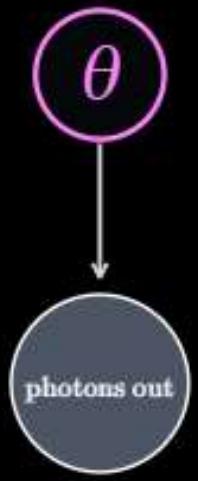
It only counted X-rays.

One beam, one block



Toy numbers.

One block



X **Photons out**

Binomial(1,000, θ)

1,000 photons, each gets through with chance θ

θ ?

chance one photon gets through



How likely is 368?



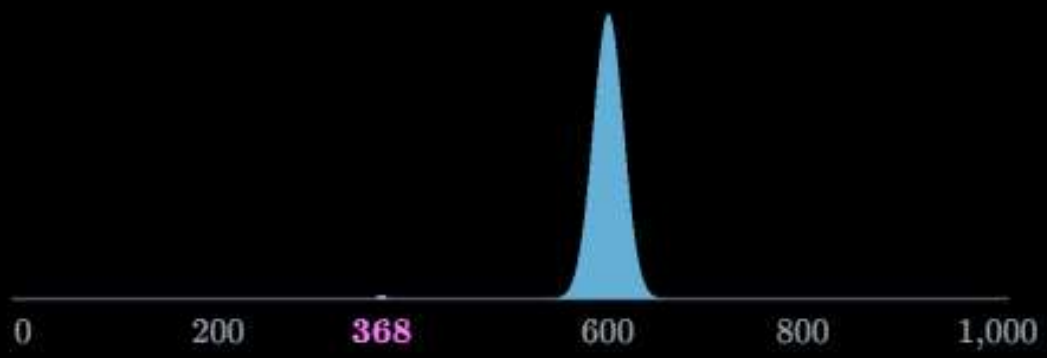
368
out

Binomial(1,000, θ)

θ **0.6**

chance one photon gets through

	likelihood	score
$\theta = 0.2$	0.000...046 (34 zeros)	**−34.34**
$\theta = 0.6$	0.000...099 (49 zeros)	**−49.01**



Toy numbers.

How likely is 368?

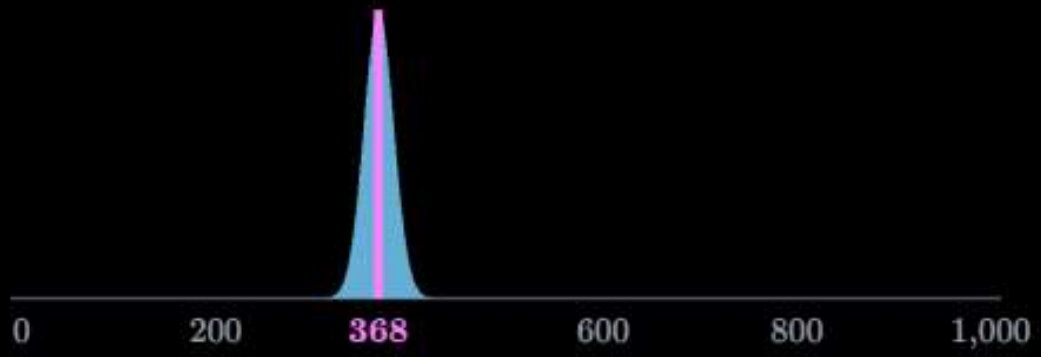


Binomial(1,000, θ) **top:** $\theta = 368 \div 1{,}000 = \mathbf{0.368}$



chance one photon gets through

	likelihood	score
$\theta = 0.2$	0.000...046 (34 zeros)	**−34.34**
$\theta = 0.6$	0.000...099 (49 zeros)	**−49.01**
$\theta = 0.4$	0.00303	**−2.52**
$\theta = 0.368$	0.0262	**−1.58**

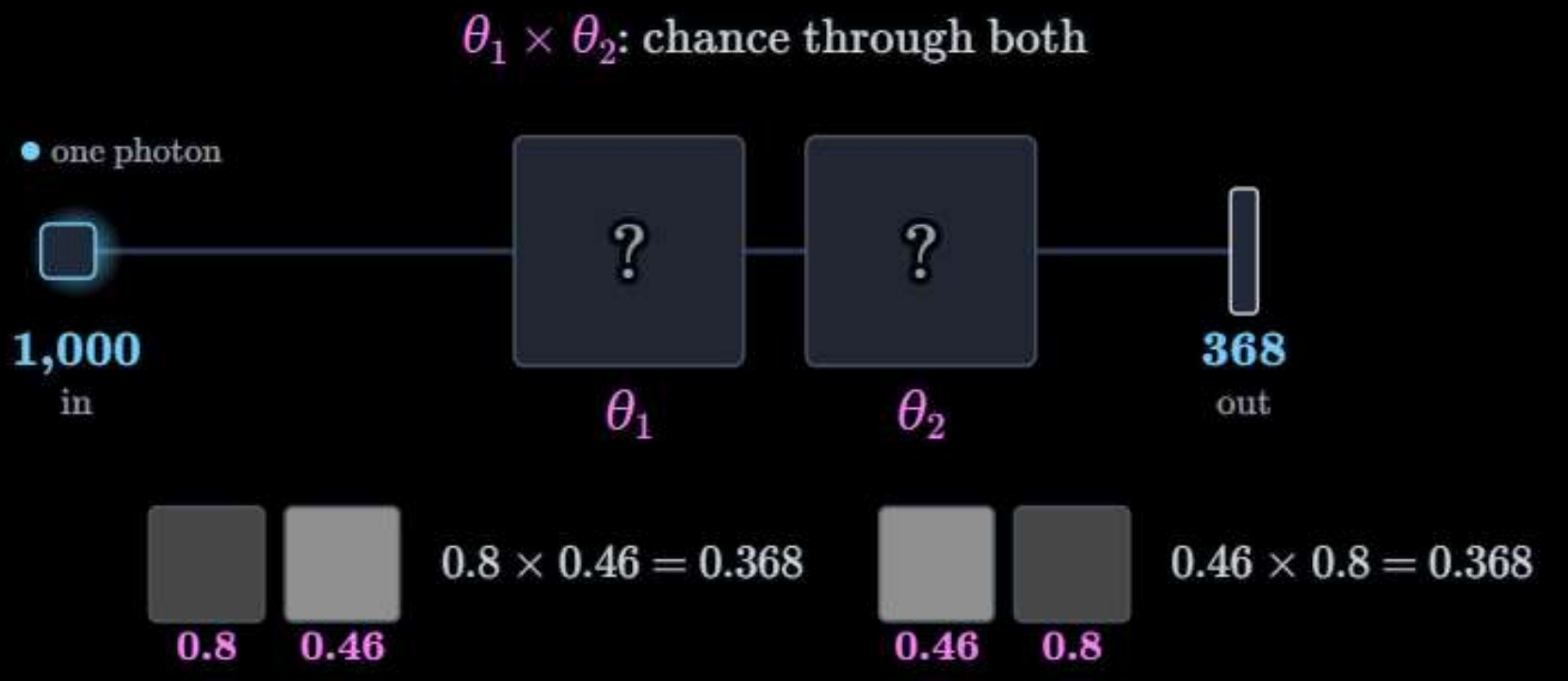




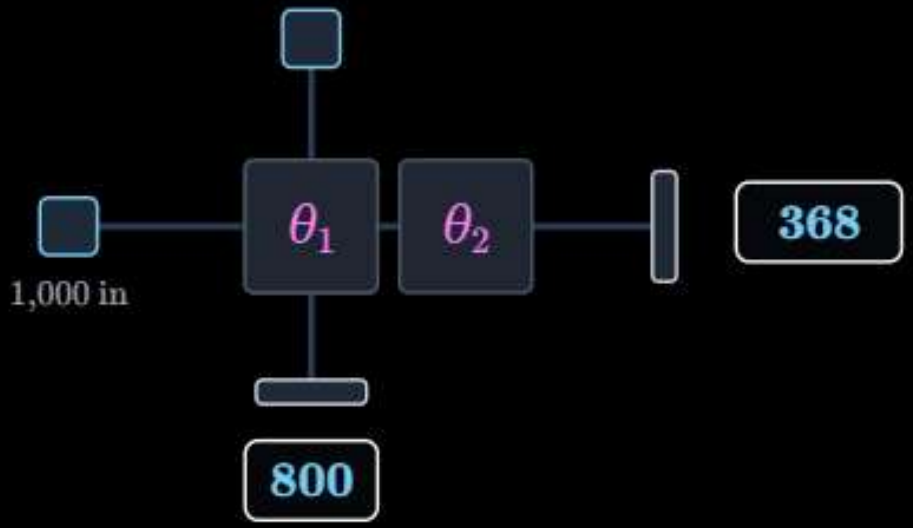
Toy numbers.

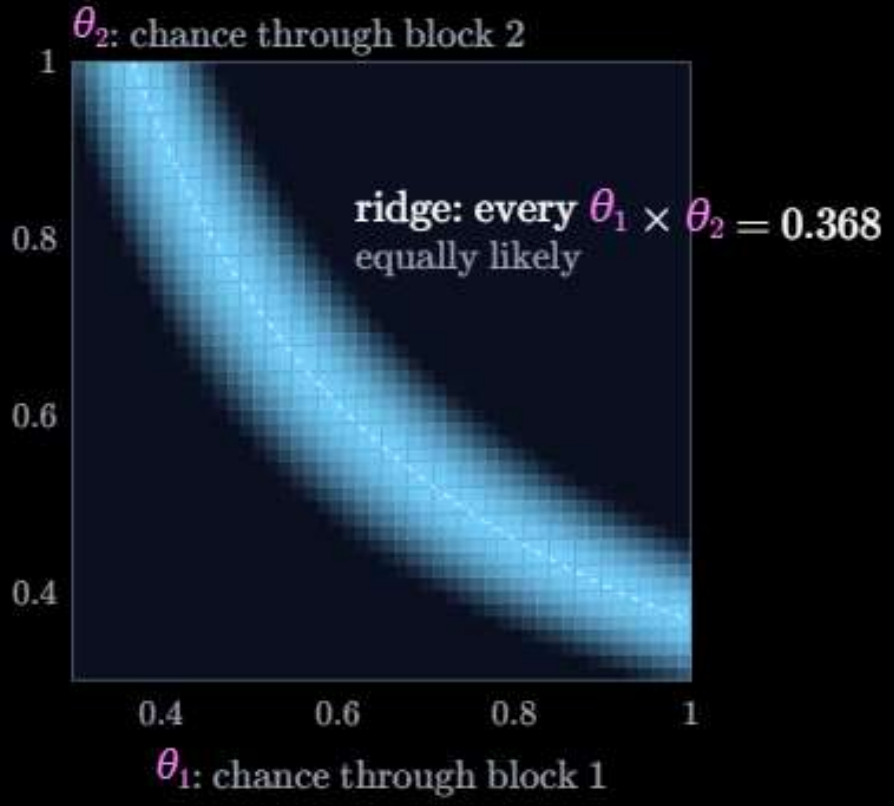
Two blocks, one beam

You guess: which one is bone?



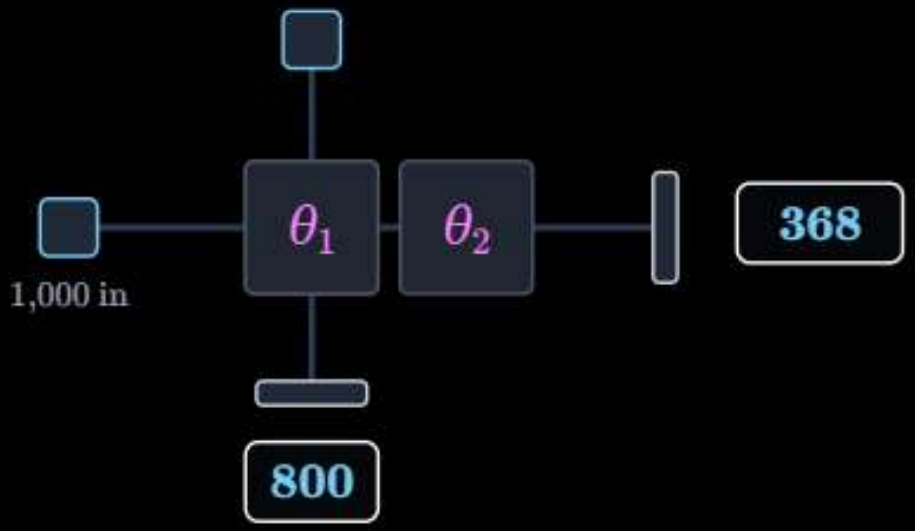
Their likelihood: a ridge





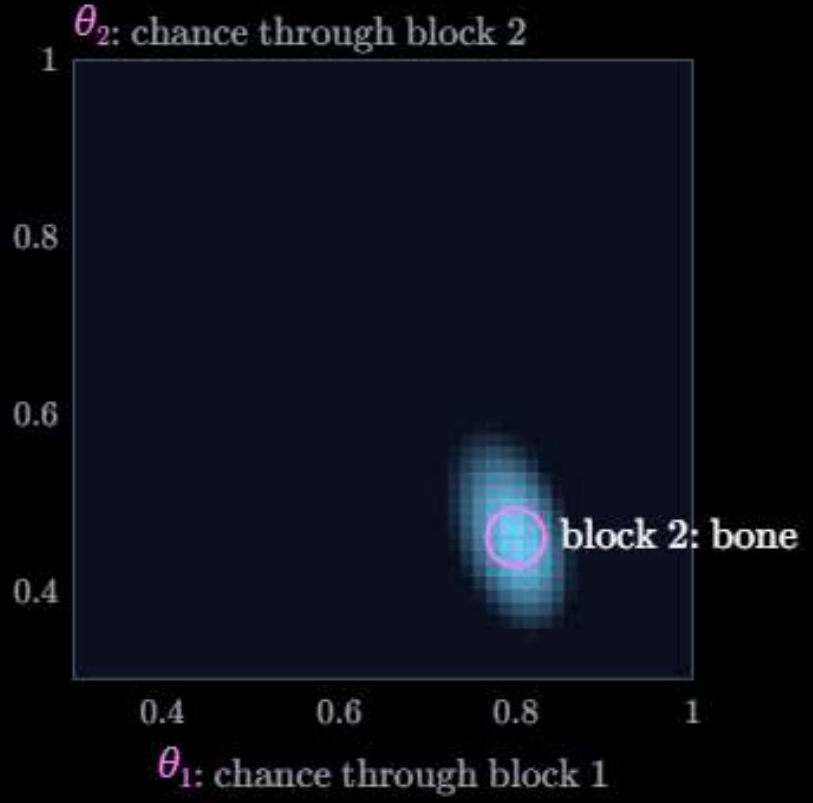
Toy numbers.

Their likelihood: a ridge



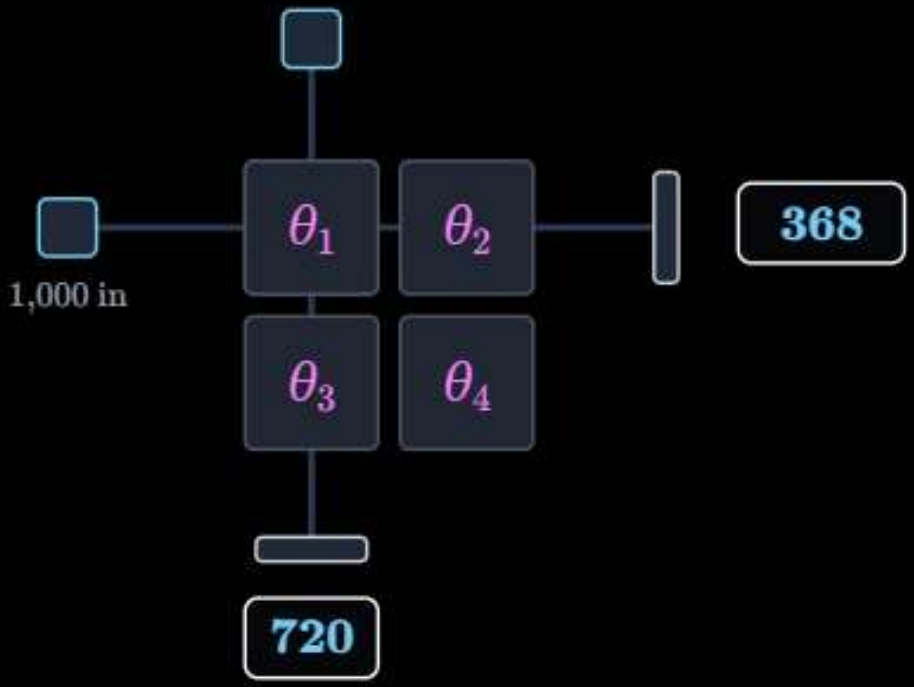
$\theta_1 = 800 \div 1{,}000 = 0.8$

$\theta_2 = 0.368 \div 0.8 = 0.46$



Toy numbers.

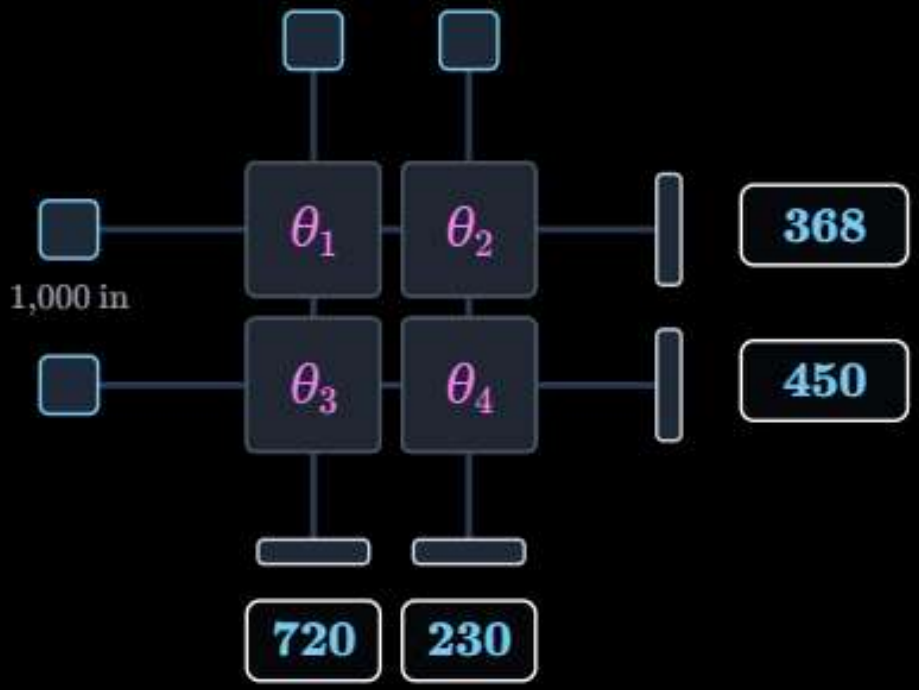
Four blocks



$\theta_1 \ldots \theta_4$: chance through each block

Toy numbers.

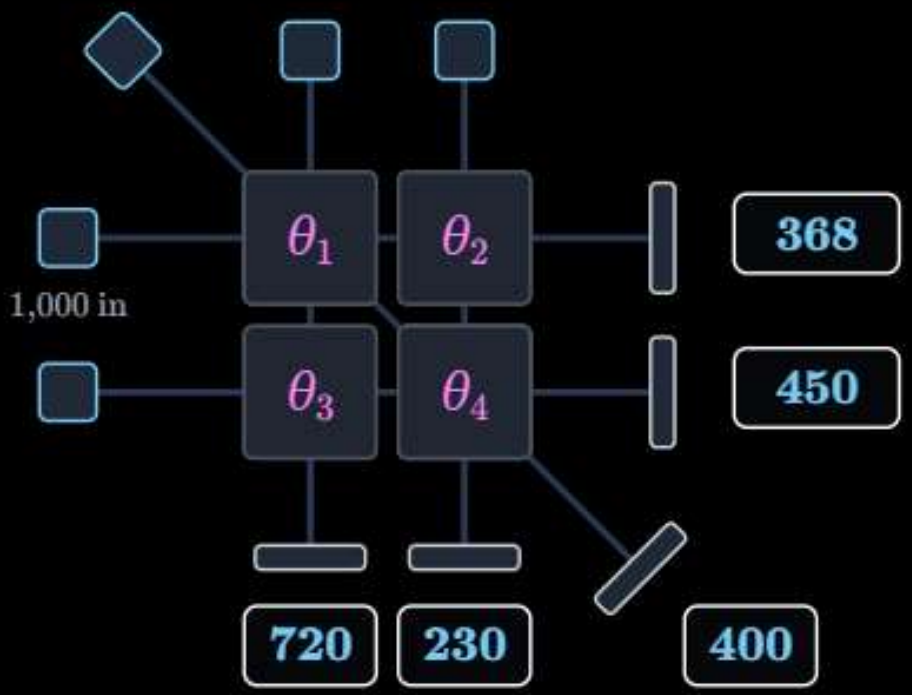
Four blocks



$\theta_1 \ldots \theta_4$: chance through each block

Toy numbers.

Four blocks



both fit all four beams

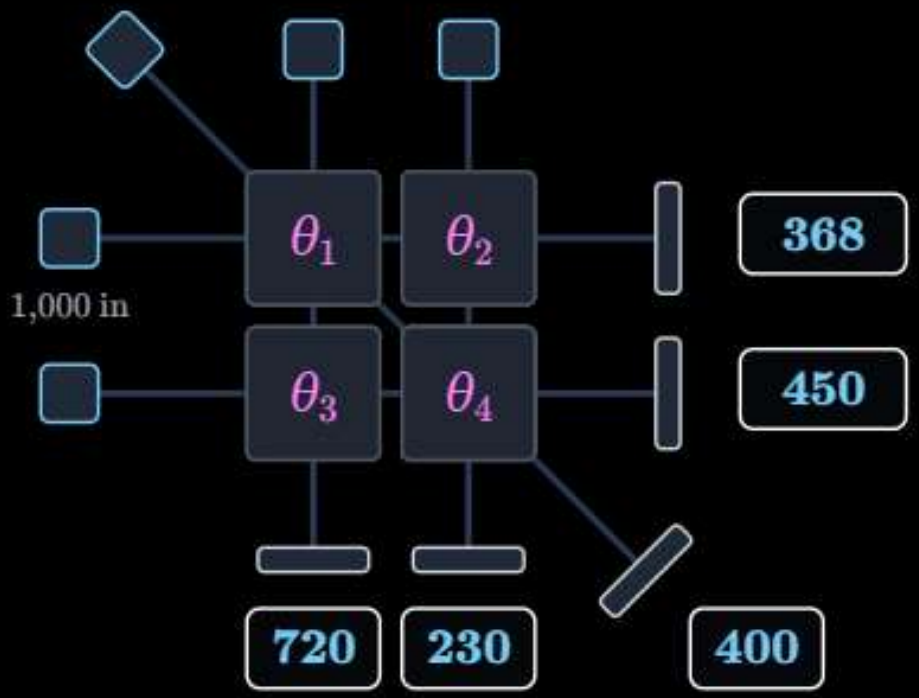
A	0.8	0.46
	0.9	0.5

B	1	0.368
	0.72	0.625

$\theta_1 \ldots \theta_4$: chance through each block

Toy numbers.

Four blocks



both fit all four beams

A

0.8	0.46
0.9	0.5

diagonal: 400

B

1	0.368
0.72	0.625

diagonal: 625

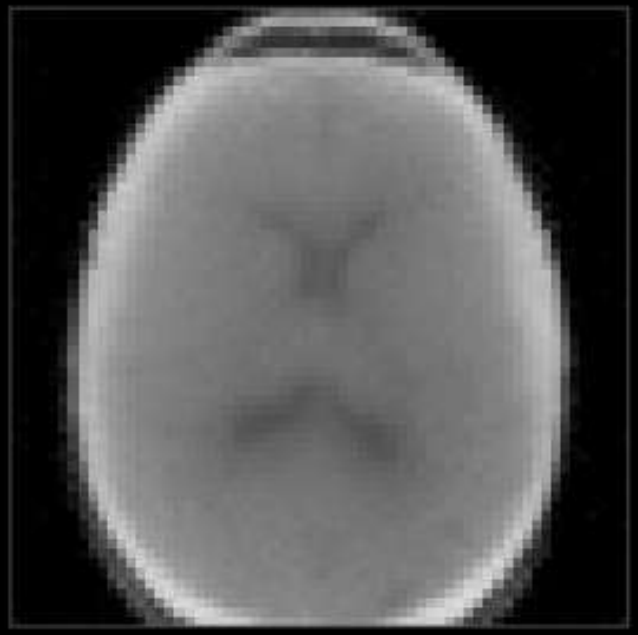
$\theta_1 \ldots \theta_4$: chance through each block

Toy numbers.

One parameter per pixel

one hop: every pixel steps uphill

hops: 10





score, higher is better



Image: our simulated scan of a real slice (M. Häggström, CC0), 64 × 64.

One parameter per pixel

one hop: every pixel steps uphill

hops: 150



score, higher is better



Image: our simulated scan of a real slice (M. Häggström, CC0), 64 × 64.

One parameter per pixel

hops: 150



score, higher
is better



the real slice

Image: our simulated scan of a real slice (M. Hāggstrōm, CC0), 64 × 64.



Photo: Warren LeMay, CC0.

Your turn: a pyramid, with muons.

One plate first, then find the hidden room.

That's all folks!



- Model first: you write the story
- Likelihood: P(observed data | θ)
- Score: log likelihood
- MLE: the θ with the highest likelihood
- Use a computer: the score, $\partial \mathrm{LL}/\partial \theta$, then gradient ascent



Now you know how it saw inside.



Image: Mikael Häggström, CC0.

The picture that makes the counts **most likely.**